MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

# Treatment of Obstetric/Gynecologic Infections

| Clinical Setting | Empiric Therapy | Duration | Comments |
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| **Pelvic Inflammatory Disease** (Endometritis, Salpingitis, Tubo-ovarian abscess, pelvic peritonitis)<br>• *Neisseria gonorrhoeae*<br>• *Chlamydia trachomatis*<br>• Anaerobes<br>• Enteric gram-negative rods<br>• *Streptococcus agalactiae*<br>• *Mycoplasma hominis*<br>• *Ureaplasma urealyticum* | Empiric Regimen<br>**Cefoxitin*** 2 g IV q6h<br>**+ Doxycycline** 200 mg PO x1, then 100 mg PO BID<br><br>Severe non-IgE mediated PCN or high-risk cephalosporin allergy<br>**Clindamycin** 900 mg IV q8h<br>**+ Gentamicin*** 5 mg/kg IV q24h<br>+ **Azithromycin** 2 g PO once<br><br>Step Down Oral Therapy<br>*With tubo-ovarian abscess:*<br>**Doxycycline** 100 mg PO BID<br>**+ Clindamycin** 450 mg PO QID<br>OR<br>**Metronidazole** 500 mg PO BID<br>*Without tubo-ovarian abscess:*<br>**Doxycycline** 100 mg PO BID | General:<br>14 days at minimum<br><br>Parenteral therapy can be switched to oral therapy 24-48 hours after clinical improvement. | • In women with tubo-ovarian abscesses, at least 24 hours of inpatient observation is recommended.<br>• Extended interval gentamicin dosing (5 mg/kg IV q24h) is preferred. Therapeutic drug monitoring may be necessary and dosing adjustments should be made with PharmD.<br>• Duration of therapy should be a minimum of 14 days without drainage of tubo-ovarian abscess. Final duration should be dependent on resolution of the abscess on follow-up imaging studies. Some patients may require 4-6 weeks of antimicrobial therapy.<br>• For *outpatient* management of mild-to-moderate PID:<br>○ **Ceftriaxone** 500 mg IM x1 dose<br>○ + **Doxycycline** 200 mg PO x1, then 100 mg PO BID<br>○ + **Metronidazole** 500 mg PO BID<br>○ Women who do not respond to IM/oral therapy within 72 hours should be reevaluated to confirm the diagnosis and should be given intravenous therapy |
| **Post-operative intra-abdominal abscess or peritonitis after gynecologic surgery**<br>• Gram-negative bacilli (*E. coli*, ok*Klebsiella*, *Proteus*, *Pseudomonas, Enterobacter*)<br>• Enteric gram-positive organisms (*Streptococcus, Enterococcus*) | Empiric Regimen:<br>**Piperacillin-tazobactam***4.5 g IV q6h<br><br>Low-High Risk PCN allergy<br>**Cefepime*** 2 g IV q8h<br>+ **Metronidazole** 500 mg PO/IV q12h<br><br>Consider the addition of vancomycin** for Enterococcus coverage in critically ill patients<br><br>Severe non-IgE beta-lactam allergy<br>**Vancomycin**** IV (see nomogram)<br>+ **Aztreonam*** 2 g IV q8h<br>+ **Metronidazole** 500 mg PO/IV q12h | General:<br>5-7 days<br><br>Duration of therapy may be extended with inadequate source control or persistent clinical symptoms or signs of infection. | • Ampicillin/sulbactam is not recommended for use because of high rates of resistance among community- acquired *E. coli.*<br>• Adjust antimicrobial coverage based on organism identification and antimicrobial susceptibilities.<br>• *Enterococcus* coverage: Empiric antimicrobial coverage for vancomycin resistant enterococcus (VRE) is not recommended except in critically ill liver transplant recipients, patients with a previous history of VRE intra-abdominal infection, or patients with septic shock who are colonized with VRE. Linezolid 600 mg PO/IV q12h can be considered empirically but should be discontinued if VRE is not identified on blood culture.<br>• MRSA coverage: Empiric MRSA antimicrobial coverage should only be provided to patients with post-operative peritonitis who are known to be colonized with MRSA and discontinued if MRSA is not recovered in culture |

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| **Chorioamnionitis**<br>• Anaerobes<br>• *Gardnerella vaginalis*<br>• Group B streptococcus<br>• *Clostridium spp.*<br>• Enterobacteriaceae<br>• *Chlamydia trachomatis*<br>• *Mycoplasma*<br>• *Ureaplasma*<br>• *S. aureus* | Preferred<br>**Piperacillin-tazobactam*** 4.5g IV q6h<br><br>Low-to-High-Risk PCN Allergy<br>**Ceftriaxone** 2g IV q24h **+ Metronidazole** 500mg PO/IV q12h<br><br>High- Risk Cephalosporin Allergy:<br>**Meropenem*** 1g IV q8hr<br><br>Severe Non-IgE Mediated Allergy:<br>**Aztreonam*** 2g IV q8h **+ Metronidazole** 500mg PO/IV q12h **+ Vancomycin**** IV | For vaginal birth, antibiotics can be discontinued after delivery<br><br>For cesarean birth, continue antibiotics for 1 additional dose after delivery<br><br>If persistently febrile, purulent amniotic fluid, or clinically ill, continue antibiotics until 24-48h afebrile | • Duration of antibiotic therapy should be extended in the setting of bacteremia (7-14 days total duration).<br>• Please follow preoperative surgical antimicrobial prophylaxis for those undergoing cesarean delivery<br>• Metronidazole has 100% oral bioavailability. PO or IV dosing acceptable<br>• For definition of low-risk or high-risk beta lactam allergies, refer to β-lactam Allergy Evaluation (βLAES) and Empiric Therapy guidance |
| **Endometritis-**<br>• Anaerobes<br>• *Gardnerella vaginalis*<br>• Group B streptococcus<br>• *Clostridium spp.*<br>• Enterobacteriaceae<br>• *Chlamydia trachomatis* | Preferred<br>**Piperacillin-tazobactam*** 4.5g IV q6h<br><br>Low-to-High-Risk PCN Allergy<br>**Ceftriaxone** 2g IV q24h **+ Metronidazole** 500mg PO/IV q12h<br><br>High- Risk Cephalosporin Allergy:<br>**Meropenem*** 1g IV q8hr<br><br>Severe Non-IgE Mediated Allergy:<br>**Aztreonam*** 2g IV q8h **+ Metronidazole** 500mg PO/IV q12h **+ Vancomycin**** IV | Continue until patient has clinically improved and afebrile for 24 hours. | • Duration of antibiotic therapy should be extended in the setting of bacteremia (7-14 days total duration).<br>• MRSA is not a common pathogen and thus addition of vancomycin is only recommended in patients with severe sepsis, septic shock, or those with persistent fevers. Vancomycin should be discontinued if MRSA is not isolated once patient is stable.<br>• Metronidazole has 100% oral bioavailability. PO or IV dosing acceptable<br>• For definition of low-risk or high-risk beta lactam allergies, refer to β-lactam Allergy Evaluation (βLAES) and Empiric Therapy guidance |

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| **Obstetric Critical Illness, Sepsis or Septic Shock** | Preferred Therapy<br>**Piperacillin-tazobactam*** 4.5 g IV q6h<br>+ **Vancomycin**** IV<br><br>Low-to-high risk PCN allergy<br>**Cefepime*** 2 g IV q8h<br>**+ Metronidazole** 500 mg PO/IV q12h<br>**+ Vancomycin**** IV (see nomogram)<br><br>High-Risk Cephalosporin Allergy<br>**Meropenem*** 2g IV q8h **+ Vancomycin**** IV<br><br>Severe non-IgE beta-lactam allergy<br>**Vancomycin**** IV (see nomogram)<br>+ **Aztreonam*** 2 g IV q8h<br>+ **Metronidazole** 500 mg PO/IV q12h | **Sepsis w/o bacteremia:**<br>7-14 days, can step-down to oral therapy when stable<br><br>**Sepsis with gram-negative bacteremia:**<br>7-14 days with IV antibiotics or oral quinolone if susceptible | • Severe infection defined by persistent fevers or tachycardia, purulent amniotic fluid, signs of end organ damage<br>• Aztreonam does not have gram-positive activity and should be used in combination with vancomycin.<br>• MRSA is not a common pathogen and thus addition of vancomycin is only recommended in patients with severe sepsis or septic shock. Vancomycin should be discontinued if MRSA is not isolated once patient is stable.<br>• Consider ID consult<br>• For definition of low-risk or high-risk beta lactam allergies, refer to β-lactam Allergy Evaluation (βLAES) and Empiric Therapy guidance |

*Dose may need to be adjusted for renal dysfunction ** For ADULTS: Dose per vancomycin nomogram

Beta-Lactam Allergy Information (For more PCN allergy information, please see *Beta-Lactam Evaluation and Empiric Therapy* guidance.)
Low risk includes remote unknown reaction, pruritis without rash, urticaria/hives with no other symptoms, and mild to severe rash with no other symptoms. Patients with penicillin allergy who have used cephalosporins in the past without reaction should receive cefazolin.
High-risk includes patients who have had any of the following reactions to a penicillin drug: 1. anaphylaxis 2. cardiovascular symptoms (syncope/passing out, arrhythmia) 3. angioedema/swelling 4. respiratory distress. Patients with a high-risk cephalosporin allergy, avoid PCNs and cephalosporins.
Severe non-IgE mediated PCN allergy includes organ damage (kidney/liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, Stevens-Johnson Syndrome/Toxic Epidermal Necrosis, rash with pustules, DRESS, serum sickness.

| **Last revised:** 04/2026 | **Antimicrobial Stewardship Subcommittee:** 04/2026 | **P&T:** 05/2026 |
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**References**

| | **Citation** |
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**Revision History**

| **Date** | **Revision** |
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| 03/2021 | Updated vancomycin dosing & hyperlinks |

*The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.*

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| 07/2021 | Updated PID outpatient recommendations |
| 01/2025 | Updated outpatient treatment dosing |
| 04/2026 | Updated peripartum antibiotic treatment regimens |

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